MONISH.S

ITA1409 – ETHICAL HACKING

ASSIGNMENT 2 – CO2

ASSIGNMENT DETAILS	INFORMATION
Student Name	MONISH.S
Course Code	ITA1409
Course Title	Ethical Hacking
Assignment	Assignment 2 – CO2
Nature	Individual
Duration	1 Week
Total Marks	50
Total Questions	7
Total Pages	70 (10 pages per question)

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 1 of 10 – Cover and strategy overview

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Authorization must name the domain owner, exact parent domain, approved subdomains, permitted source types, collection dates and assessor. A related hostname discovered during collection is not automatically in scope. Third-party platforms require separate ownership and permission review.

For the section 'Cover and strategy overview', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 2 of 10 – Requirements, authorization and scope

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Passive methods include public WHOIS/RDAP, search-engine results, archived pages and public documents. Active methods include direct DNS queries to an approved resolver or authoritative server. Classification depends on interaction with the target, not simply on the tool selected.

For the section 'Requirements, authorization and scope', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

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Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 3 of 10 – Passive and active techniques

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

A well-protected target is predicted to expose limited public technical detail. An information-leaking target may have public documents or old hostnames. A misconfigured DNS target may expose extra records in a controlled lab. A minimal-footprint target may publish only essential records. All observations should be labelled simulated unless collected in an authorized lab.

For the section 'Passive and active techniques', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 4 of 10 – Multi-source collection plan

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Authorization must name the domain owner, exact parent domain, approved subdomains, permitted source types, collection dates and assessor. A related hostname discovered during collection is not automatically in scope. Third-party platforms require separate ownership and permission review.

For the section 'Multi-source collection plan', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 5 of 10 – Four contrasting target profiles

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Passive methods include public WHOIS/RDAP, search-engine results, archived pages and public documents. Active methods include direct DNS queries to an approved resolver or authoritative server. Classification depends on interaction with the target, not simply on the tool selected.

For the section 'Four contrasting target profiles', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
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Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 6 of 10 – Prediction versus observation

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

A well-protected target is predicted to expose limited public technical detail. An information-leaking target may have public documents or old hostnames. A misconfigured DNS target may expose extra records in a controlled lab. A minimal-footprint target may publish only essential records. All observations should be labelled simulated unless collected in an authorized lab.

For the section 'Prediction versus observation', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 7 of 10 – Reliability, contradictions and freshness

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Authorization must name the domain owner, exact parent domain, approved subdomains, permitted source types, collection dates and assessor. A related hostname discovered during collection is not automatically in scope. Third-party platforms require separate ownership and permission review.

For the section 'Reliability, contradictions and freshness', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 8 of 10 – Out-of-scope data and privacy

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Passive methods include public WHOIS/RDAP, search-engine results, archived pages and public documents. Active methods include direct DNS queries to an approved resolver or authoritative server. Classification depends on interaction with the target, not simply on the tool selected.

For the section 'Out-of-scope data and privacy', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 9 of 10 – Scaling to fifty subdomains

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

A well-protected target is predicted to expose limited public technical detail. An information-leaking target may have public documents or old hostnames. A misconfigured DNS target may expose extra records in a controlled lab. A minimal-footprint target may publish only essential records. All observations should be labelled simulated unless collected in an authorized lab.

For the section 'Scaling to fifty subdomains', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q1 – EVALUATE AND ENGINEER A FOOTPRINTING STRATEGY

Page 10 of 10 – Final strategy and judgement

This question designs a footprinting strategy for an authorized organization represented by the reserved example domain example-corp.example. The domain is used only as a documentation example; no live organization is scanned. A sound strategy combines WHOIS/RDAP, DNS, search engines and public documents because each source provides a different view of the public attack surface. WHOIS may show registrar and registration details, DNS may reveal published service records, search engines may expose indexed pages, and archives may preserve historical content.

Authorization must name the domain owner, exact parent domain, approved subdomains, permitted source types, collection dates and assessor. A related hostname discovered during collection is not automatically in scope. Third-party platforms require separate ownership and permission review.

For the section 'Final strategy and judgement', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Source	Method	Expected information	Caution
WHOIS/RDAP	Passive	Registrar, dates, NS references	Privacy and delay
DNS A/MX/NS/TXT	Active query	Address and service records	Resolver and TTL
Search engine	Passive	Indexed pages and documents	Index may be stale
Archive	Passive	Historical content	Not proof of current service

Process representation

Authorization → Scope → Passive sources → Approved DNS → Validate → Analyse → Report

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 1 of 10 – DNS objectives and boundaries

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

The workflow records the exact domain, query type, resolver, timestamp, response status, TTL and answer. Example command patterns for an authorized lab are: dig example-corp.example A; dig example-corp.example MX; dig example-corp.example NS; dig example-corp.example TXT; and nslookup -type=MX example-corp.example.

For the section 'DNS objectives and boundaries', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 2 of 10 – DNS record types and interpretation

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

A zone transfer can reveal hostnames and relationships across a DNS zone. It must be attempted only when the written scope explicitly permits it. A successful transfer is evidence of information exposure, not proof that every listed host is vulnerable. Restrict transfers to approved secondary servers and use TSIG where appropriate.

For the section 'DNS record types and interpretation', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 3 of 10 – dig and nslookup workflow

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

Split-horizon DNS may return different internal and external answers. Delegation may place a subdomain under a separate DNS operator. Caching can cause stale responses until TTL expiry. Record the resolver and query time, compare approved sources and avoid interpreting one response as a complete view of the DNS environment.

For the section 'dig and nslookup workflow', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 4 of 10 – Six contrasting DNS cases

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

The workflow records the exact domain, query type, resolver, timestamp, response status, TTL and answer. Example command patterns for an authorized lab are: dig example-corp.example A; dig example-corp.example MX; dig example-corp.example NS; dig example-corp.example TXT; and nslookup -type=MX example-corp.example.

For the section 'Six contrasting DNS cases', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 5 of 10 – Zone-transfer assessment

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

A zone transfer can reveal hostnames and relationships across a DNS zone. It must be attempted only when the written scope explicitly permits it. A successful transfer is evidence of information exposure, not proof that every listed host is vulnerable. Restrict transfers to approved secondary servers and use TSIG where appropriate.

For the section 'Zone-transfer assessment', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 6 of 10 – Split-horizon, delegation and caching

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

Split-horizon DNS may return different internal and external answers. Delegation may place a subdomain under a separate DNS operator. Caching can cause stale responses until TTL expiry. Record the resolver and query time, compare approved sources and avoid interpreting one response as a complete view of the DNS environment.

For the section 'Split-horizon, delegation and caching', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 7 of 10 – Prediction versus observation

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

The workflow records the exact domain, query type, resolver, timestamp, response status, TTL and answer. Example command patterns for an authorized lab are: dig example-corp.example A; dig example-corp.example MX; dig example-corp.example NS; dig example-corp.example TXT; and nslookup -type=MX example-corp.example.

For the section 'Prediction versus observation', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 8 of 10 – Restricting transfers and TSIG

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

A zone transfer can reveal hostnames and relationships across a DNS zone. It must be attempted only when the written scope explicitly permits it. A successful transfer is evidence of information exposure, not proof that every listed host is vulnerable. Restrict transfers to approved secondary servers and use TSIG where appropriate.

For the section 'Restricting transfers and TSIG', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 9 of 10 – Redesigned workflow and validation

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

Split-horizon DNS may return different internal and external answers. Delegation may place a subdomain under a separate DNS operator. Caching can cause stale responses until TTL expiry. Record the resolver and query time, compare approved sources and avoid interpreting one response as a complete view of the DNS environment.

For the section 'Redesigned workflow and validation', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q2 – BUILD A RELIABLE DNS FOOTPRINTING WORKFLOW

Page 10 of 10 – Final DNS workflow and conclusion

DNS footprinting maps domain names to network and service information. The required workflow retrieves A, MX, NS and TXT records and considers a zone transfer only when explicit written permission exists. A record commonly identifies an IPv4 address, MX identifies mail exchangers, NS identifies authoritative name servers and TXT contains text data such as verification or email-security policy. Resolver choice, TTL, caching and delegation can affect what an external tester sees.

The workflow records the exact domain, query type, resolver, timestamp, response status, TTL and answer. Example command patterns for an authorized lab are: dig example-corp.example A; dig example-corp.example MX; dig example-corp.example NS; dig example-corp.example TXT; and nslookup -type=MX example-corp.example.

For the section 'Final DNS workflow and conclusion', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Record	Purpose	Interpretation caution
A	IPv4 mapping	May vary by resolver
MX	Mail routing	Priority and provider matter
NS	Authoritative servers	Check delegation
TXT	Text / policy	May have multiple values
AXFR	Zone transfer	Only with written approval

Process representation

Confirm scope → Query A/MX/NS/TXT → Interpret → Authorized AXFR if approved → Countermeasure → Retest

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 1 of 10 – Utility purpose and ethical boundary

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

A job advertisement may reveal technology or team responsibilities; an annual report may describe departments and suppliers; a press release may reveal projects; a public profile may disclose professional roles. The report explains how the information could be misused without converting it into a ready-to-send social-engineering message.

For the section 'Utility purpose and ethical boundary', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 2 of 10 – Public-source inventory

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Phishing may use project context or familiar terminology. Vishing may exploit publicly visible job roles or office contact information. Pretexting may rely on knowledge of departments or suppliers. Tailgating risk may increase if public material reveals access arrangements. The defensive response is verification, reporting, badge control and employee awareness.

For the section 'Public-source inventory', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 3 of 10 – Search operators and query design

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Test organizations are fictional: a rich profile contains many current public facts; a sparse profile contains little information; a misleading profile contains old or contradictory facts. Predict expected human-risk exposure before collection, then compare the simulated observations and identify which sources provide reliable, actionable context.

For the section 'Search operators and query design', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 4 of 10 – Public information item register

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

A job advertisement may reveal technology or team responsibilities; an annual report may describe departments and suppliers; a press release may reveal projects; a public profile may disclose professional roles. The report explains how the information could be misused without converting it into a ready-to-send social-engineering message.

For the section 'Public information item register', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 5 of 10 – Phishing and vishing risk mapping

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Phishing may use project context or familiar terminology. Vishing may exploit publicly visible job roles or office contact information. Pretexting may rely on knowledge of departments or suppliers. Tailgating risk may increase if public material reveals access arrangements. The defensive response is verification, reporting, badge control and employee awareness.

For the section 'Phishing and vishing risk mapping', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 6 of 10 – Pretexting and tailgating mapping

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Test organizations are fictional: a rich profile contains many current public facts; a sparse profile contains little information; a misleading profile contains old or contradictory facts. Predict expected human-risk exposure before collection, then compare the simulated observations and identify which sources provide reliable, actionable context.

For the section 'Pretexting and tailgating mapping', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 7 of 10 – Rich, sparse and misleading organizations

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

A job advertisement may reveal technology or team responsibilities; an annual report may describe departments and suppliers; a press release may reveal projects; a public profile may disclose professional roles. The report explains how the information could be misused without converting it into a ready-to-send social-engineering message.

For the section 'Rich, sparse and misleading organizations', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 8 of 10 – Prediction and source-risk analysis

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Phishing may use project context or familiar terminology. Vishing may exploit publicly visible job roles or office contact information. Pretexting may rely on knowledge of departments or suppliers. Tailgating risk may increase if public material reveals access arrangements. The defensive response is verification, reporting, badge control and employee awareness.

For the section 'Prediction and source-risk analysis', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 9 of 10 – Additional sources and countermeasures

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

Test organizations are fictional: a rich profile contains many current public facts; a sparse profile contains little information; a misleading profile contains old or contradictory facts. Predict expected human-risk exposure before collection, then compare the simulated observations and identify which sources provide reliable, actionable context.

For the section 'Additional sources and countermeasures', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q3 – CREATE A COMPETITIVE-INTELLIGENCE AWARENESS UTILITY

Page 10 of 10 – Utility evaluation and recommendations

The competitive-intelligence utility collects public business information from job advertisements, annual reports, press releases and public profiles. Its purpose is defensive awareness, not an attack against real people. No phishing message is sent, no employee is called and no identity is impersonated. Each public item is recorded with its source, date, summary, reliability and possible human-risk implication. Search operators such as site:, filetype: and intitle: help document reproducible public searches.

A job advertisement may reveal technology or team responsibilities; an annual report may describe departments and suppliers; a press release may reveal projects; a public profile may disclose professional roles. The report explains how the information could be misused without converting it into a ready-to-send social-engineering message.

For the section 'Utility evaluation and recommendations', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Public item	Potential technique	Defensive indicator
Job advertisement	Pretexting / phishing context	Verify requests independently
Annual report	Pretexting context	Do not trust unexpected authority
Press release	Phishing context	Check links and attachments
Public profile	Vishing context	Use known callback numbers
Access details	Tailgating context	Enforce visitor and badge rules

Process representation

Public source → Record provenance → Map human risk → Identify warning sign → Awareness control

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 1 of 10 – Validation problem and objectives

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Scope is checked first because out-of-scope items must not be enriched. Freshness is then assessed against a documented period appropriate to the record type. Values are normalized before duplicate detection so that case differences or trailing dots do not create separate records. Source reliability is evaluated using authority, directness, date and reproducibility.

For the section 'Validation problem and objectives', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 2 of 10 – Validation fields and data model

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Use at least five test sequences: all-valid records; one out-of-scope item; one stale item; duplicate values with different formatting; and an unverifiable source. Predict the expected accepted, rejected and re-check counts before evaluating the input. Keep a reason code for every decision.

For the section 'Validation fields and data model', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 3 of 10 – Scope validation and rejection

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

A re-check item is not counted as accepted until a second source or owner confirmation supports it. Rejected records remain in an audit log but are excluded from the verified footprint. As volume grows to fifty targets, batch normalization and structured reporting reduce effort while human review remains necessary for scope changes.

For the section 'Scope validation and rejection', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 4 of 10 – Freshness and source reliability

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Scope is checked first because out-of-scope items must not be enriched. Freshness is then assessed against a documented period appropriate to the record type. Values are normalized before duplicate detection so that case differences or trailing dots do not create separate records. Source reliability is evaluated using authority, directness, date and reproducibility.

For the section 'Freshness and source reliability', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 5 of 10 – Duplicate detection and normalization

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Use at least five test sequences: all-valid records; one out-of-scope item; one stale item; duplicate values with different formatting; and an unverifiable source. Predict the expected accepted, rejected and re-check counts before evaluating the input. Keep a reason code for every decision.

For the section 'Duplicate detection and normalization', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 6 of 10 – Five adversarial test sequences

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

A re-check item is not counted as accepted until a second source or owner confirmation supports it. Rejected records remain in an audit log but are excluded from the verified footprint. As volume grows to fifty targets, batch normalization and structured reporting reduce effort while human review remains necessary for scope changes.

For the section 'Five adversarial test sequences', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 7 of 10 – Pass, fail and re-check predictions

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Scope is checked first because out-of-scope items must not be enriched. Freshness is then assessed against a documented period appropriate to the record type. Values are normalized before duplicate detection so that case differences or trailing dots do not create separate records. Source reliability is evaluated using authority, directness, date and reproducibility.

For the section 'Pass, fail and re-check predictions', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 8 of 10 – Validation order and effort

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Use at least five test sequences: all-valid records; one out-of-scope item; one stale item; duplicate values with different formatting; and an unverifiable source. Predict the expected accepted, rejected and re-check counts before evaluating the input. Keep a reason code for every decision.

For the section 'Validation order and effort', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 9 of 10 – Scaling to fifty targets

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

A re-check item is not counted as accepted until a second source or owner confirmation supports it. Rejected records remain in an audit log but are excluded from the verified footprint. As volume grows to fifty targets, batch normalization and structured reporting reduce effort while human review remains necessary for scope changes.

For the section 'Scaling to fifty targets', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q4 – DESIGN A RECONNAISSANCE DATA-VALIDATION MECHANISM

Page 10 of 10 – Redesigned validation and conclusion

Reconnaissance data can be incomplete, stale, duplicated, unreliable or outside the approved scope. The validation mechanism checks each item before it is used. A record should include an item ID, normalized value, source, collection time, scope decision, freshness status, duplicate key, confidence and final status. The outcomes are accepted, rejected or re-check. Out-of-scope information is rejected early; uncertain information is held until it can be confirmed.

Scope is checked first because out-of-scope items must not be enriched. Freshness is then assessed against a documented period appropriate to the record type. Values are normalized before duplicate detection so that case differences or trailing dots do not create separate records. Source reliability is evaluated using authority, directness, date and reproducibility.

For the section 'Redesigned validation and conclusion', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Check	Pass condition	Failure action
Scope	Exact item is approved	Reject
Freshness	Within defined period	Re-check
Duplicate	Unique after normalization	Link to canonical item
Reliability	Traceable credible source	Re-check / reject

Process representation

Normalize → Scope check → Freshness → Duplicate check → Reliability → Accept / Re-check / Reject

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 1 of 10 – Hierarchical model and boundaries

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A node record contains node ID, parent ID, name, node type, source, discovery date, scope status and terminal flag. The root is the approved parent domain. Children may be subdomains, departmental portals or named services. A third-party service can be recorded as a relationship without granting permission to test the provider.

For the section 'Hierarchical model and boundaries', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 2 of 10 – Domain, subdomain and service nodes

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A traversal uses a queue or stack. It begins at the approved root, checks the scope flag before processing, records the node and adds only approved children. It stops when a node is terminal, out of scope or has no discovered children. The log records the number of visited nodes and the reason each branch ended.

For the section 'Domain, subdomain and service nodes', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 3 of 10 – Traversal algorithm and stopping rules

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

Test a shallow tree, a deep chain and an irregular tree with branches of different sizes. Include an empty branch and an excluded third-party service. Predict node count and depth before traversal, then compare actual visited nodes. Correctness means every approved node is visited once and no excluded node is processed.

For the section 'Traversal algorithm and stopping rules', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 4 of 10 – Shallow footprint test

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A node record contains node ID, parent ID, name, node type, source, discovery date, scope status and terminal flag. The root is the approved parent domain. Children may be subdomains, departmental portals or named services. A third-party service can be recorded as a relationship without granting permission to test the provider.

For the section 'Shallow footprint test', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 5 of 10 – Deeply nested footprint test

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A traversal uses a queue or stack. It begins at the approved root, checks the scope flag before processing, records the node and adds only approved children. It stops when a node is terminal, out of scope or has no discovered children. The log records the number of visited nodes and the reason each branch ended.

For the section 'Deeply nested footprint test', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 6 of 10 – Irregular branches and empty nodes

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

Test a shallow tree, a deep chain and an irregular tree with branches of different sizes. Include an empty branch and an excluded third-party service. Predict node count and depth before traversal, then compare actual visited nodes. Correctness means every approved node is visited once and no excluded node is processed.

For the section 'Irregular branches and empty nodes', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 7 of 10 – Predicted node counts and evidence

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A node record contains node ID, parent ID, name, node type, source, discovery date, scope status and terminal flag. The root is the approved parent domain. Children may be subdomains, departmental portals or named services. A third-party service can be recorded as a relationship without granting permission to test the provider.

For the section 'Predicted node counts and evidence', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 8 of 10 – Depth, branching and effort

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A traversal uses a queue or stack. It begins at the approved root, checks the scope flag before processing, records the node and adds only approved children. It stops when a node is terminal, out of scope or has no discovered children. The log records the number of visited nodes and the reason each branch ended.

For the section 'Depth, branching and effort', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 9 of 10 – Systematic versus ad-hoc traversal

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

Test a shallow tree, a deep chain and an irregular tree with branches of different sizes. Include an empty branch and an excluded third-party service. Predict node count and depth before traversal, then compare actual visited nodes. Correctness means every approved node is visited once and no excluded node is processed.

For the section 'Systematic versus ad-hoc traversal', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q5 – EVALUATE FOOTPRINTING OF A HIERARCHICAL ORGANIZATION

Page 10 of 10 – Final traversal design and defence

The organization's public footprint is represented as a hierarchy with an authorized parent domain, subdomains, departments and services. Each node has a unique ID, parent ID, node type, source, scope status and terminal flag. A traversal visits only approved nodes and stops at out-of-scope or terminal nodes. Empty branches are recorded as empty rather than silently omitted. This design helps measure coverage and prevents uncontrolled discovery.

A node record contains node ID, parent ID, name, node type, source, discovery date, scope status and terminal flag. The root is the approved parent domain. Children may be subdomains, departmental portals or named services. A third-party service can be recorded as a relationship without granting permission to test the provider.

For the section 'Final traversal design and defence', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Field	Meaning	Example
Node ID	Unique identifier	N-001
Parent ID	Parent node	N-ROOT
Node type	Domain/subdomain/service	Subdomain
Scope	Approved/excluded/pending	Approved
Terminal	Stop traversal	Yes / No

Process representation

Approved root → Visit node → Check boundary → Record → Add approved children → Stop condition

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 1 of 10 – Evaluation objectives and constraints

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Approach A uses WHOIS/RDAP, search engines, archived pages and public documents. Approach B uses explicitly approved DNS queries and low-impact service checks if named in the engagement. The report compares outputs, query counts, unique items, duplicates, stale items, scope exceptions and elapsed time.

For the section 'Evaluation objectives and constraints', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 2 of 10 – Passive open-source approach

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Use four increasing scope sizes, for example 1, 5, 20 and 50 approved names. These are illustrative test sizes, not real measurements. At each size, record the steps required and the number of unique verified items. Identify the point where manual passive review or repeated active queries becomes impractical.

For the section 'Passive open-source approach', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 3 of 10 – Active DNS footprinting approach

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

The faster approach is not automatically better. Passive data may be quiet but old; active data may be current but creates traffic and requires tighter authorization. A balanced recommendation uses passive sources for broad context and focused active validation for approved names, with rate limits, deduplication and stop conditions.

For the section 'Active DNS footprinting approach', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 4 of 10 – Equivalent target and conditions

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Approach A uses WHOIS/RDAP, search engines, archived pages and public documents. Approach B uses explicitly approved DNS queries and low-impact service checks if named in the engagement. The report compares outputs, query counts, unique items, duplicates, stale items, scope exceptions and elapsed time.

For the section 'Equivalent target and conditions', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 5 of 10 – Four increasing scope sizes

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Use four increasing scope sizes, for example 1, 5, 20 and 50 approved names. These are illustrative test sizes, not real measurements. At each size, record the steps required and the number of unique verified items. Identify the point where manual passive review or repeated active queries becomes impractical.

For the section 'Four increasing scope sizes', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 6 of 10 – Queries, time and coverage measures

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

The faster approach is not automatically better. Passive data may be quiet but old; active data may be current but creates traffic and requires tighter authorization. A balanced recommendation uses passive sources for broad context and focused active validation for approved names, with rate limits, deduplication and stop conditions.

For the section 'Queries, time and coverage measures', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 7 of 10 – Redundancy analysis

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Approach A uses WHOIS/RDAP, search engines, archived pages and public documents. Approach B uses explicitly approved DNS queries and low-impact service checks if named in the engagement. The report compares outputs, query counts, unique items, duplicates, stale items, scope exceptions and elapsed time.

For the section 'Redundancy analysis', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 8 of 10 – Speed, completeness and detection trade-offs

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Use four increasing scope sizes, for example 1, 5, 20 and 50 approved names. These are illustrative test sizes, not real measurements. At each size, record the steps required and the number of unique verified items. Identify the point where manual passive review or repeated active queries becomes impractical.

For the section 'Speed, completeness and detection trade-offs', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 9 of 10 – Small- and large-scope recommendations

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

The faster approach is not automatically better. Passive data may be quiet but old; active data may be current but creates traffic and requires tighter authorization. A balanced recommendation uses passive sources for broad context and focused active validation for approved names, with rate limits, deduplication and stop conditions.

For the section 'Small- and large-scope recommendations', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q6 – EVALUATE FOOTPRINTING APPROACHES UNDER CONSTRAINTS

Page 10 of 10 – Final comparison and judgement

The assignment compares passive open-source collection with active DNS and service footprinting. Passive collection uses already published information and usually creates little direct interaction with the target, but may be stale. Active queries can provide more current technical observations but must be explicitly authorized and may be visible in logs. Both approaches must use the same target, scope and collection window for a fair comparison.

Approach A uses WHOIS/RDAP, search engines, archived pages and public documents. Approach B uses explicitly approved DNS queries and low-impact service checks if named in the engagement. The report compares outputs, query counts, unique items, duplicates, stale items, scope exceptions and elapsed time.

For the section 'Final comparison and judgement', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Metric	Passive OSINT	Approved DNS
Interaction	Usually none	Queries sent
Output	Context/history	DNS observations
Freshness	Variable	Resolver/cache dependent
Scope risk	Lower direct interaction	Requires exact names
Best use	Broad context	Focused verification

Process representation

Same target → Passive workflow → Approved active workflow → Measure → Compare → Recommend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 1 of 10 – Selected workflow and review context

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The original validation sequence checks source and freshness before scope and does not normalize before duplicate detection. The redesigned sequence checks scope first, normalizes the value, detects duplicates, checks freshness and then assesses source reliability. This order avoids spending effort on excluded data and reduces duplicate processing.

For the section 'Selected workflow and review context', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 2 of 10 – Diagnose limitation using evidence

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

Evaluation criteria include zero out-of-scope enrichment, valid-item coverage, number of unnecessary checks, duplicate count, re-check count and auditability. The original and redesigned workflows use the same fictional input records, freshness threshold and source-quality rules. Record predictions before calculating outcomes.

For the section 'Diagnose limitation using evidence', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 3 of 10 – Root-cause analysis

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The redesign is preferred when a dataset contains many out-of-scope or repeated items. The original may remain adequate for a small dataset that has already been scope-filtered. The trade-off is that early scope rejection depends on a current and accurate scope register. The final choice is based on measured evidence, not preference.

For the section 'Root-cause analysis', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 4 of 10 – Measurable evaluation criteria

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The original validation sequence checks source and freshness before scope and does not normalize before duplicate detection. The redesigned sequence checks scope first, normalizes the value, detects duplicates, checks freshness and then assesses source reliability. This order avoids spending effort on excluded data and reduces duplicate processing.

For the section 'Measurable evaluation criteria', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 5 of 10 – Original workflow and redesign logic

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

Evaluation criteria include zero out-of-scope enrichment, valid-item coverage, number of unnecessary checks, duplicate count, re-check count and auditability. The original and redesigned workflows use the same fictional input records, freshness threshold and source-quality rules. Record predictions before calculating outcomes.

For the section 'Original workflow and redesign logic', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 6 of 10 – Redesigned workflow and trade-offs

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The redesign is preferred when a dataset contains many out-of-scope or repeated items. The original may remain adequate for a small dataset that has already been scope-filtered. The trade-off is that early scope rejection depends on a current and accurate scope register. The final choice is based on measured evidence, not preference.

For the section 'Redesigned workflow and trade-offs', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 7 of 10 – Equivalent test conditions

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The original validation sequence checks source and freshness before scope and does not normalize before duplicate detection. The redesigned sequence checks scope first, normalizes the value, detects duplicates, checks freshness and then assesses source reliability. This order avoids spending effort on excluded data and reduces duplicate processing.

For the section 'Equivalent test conditions', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 8 of 10 – Quantitative comparison

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

Evaluation criteria include zero out-of-scope enrichment, valid-item coverage, number of unnecessary checks, duplicate count, re-check count and auditability. The original and redesigned workflows use the same fictional input records, freshness threshold and source-quality rules. Record predictions before calculating outcomes.

For the section 'Quantitative comparison', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 9 of 10 – Final judgement and limitations

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The redesign is preferred when a dataset contains many out-of-scope or repeated items. The original may remain adequate for a small dataset that has already been scope-filtered. The trade-off is that early scope rejection depends on a current and accurate scope register. The final choice is based on measured evidence, not preference.

For the section 'Final judgement and limitations', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
Coverage	Valid items accepted	No reduction
Efficiency	Unnecessary checks	Fewer than original
Duplicates	Duplicate records retained	One canonical record
Audit	Reason recorded	All outcomes traceable

Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Q7 – INTEGRATED REVIEW: DIAGNOSE, REDESIGN AND DEFEND

Page 10 of 10 – Integrated validation and conclusion

Question 7 selects the Q4 data-validation workflow for diagnosis and redesign. A weak validation order can spend effort enriching out-of-scope items and can count duplicate hostnames more than once. The redesign checks scope first and normalizes values before duplicate detection. Both workflows are compared using identical fictional inputs and measurable criteria such as scope compliance, coverage, duplicate count, re-check count and unnecessary processing steps.

The original validation sequence checks source and freshness before scope and does not normalize before duplicate detection. The redesigned sequence checks scope first, normalizes the value, detects duplicates, checks freshness and then assesses source reliability. This order avoids spending effort on excluded data and reduces duplicate processing.

For the section 'Integrated validation and conclusion', the assessor states the question being answered, identifies the evidence required, selects a method within the authorized boundary and records the decision rule. The record should be independently reviewable so another assessor can repeat the documented step in the approved lab.

Evidence should include an engagement or test-case identifier, approved target, date and time, source or command, observed output, interpretation, scope decision and reviewer note. Raw output is stored separately from the report summary. Sensitive values are redacted, and real credentials or personal records are not included in illustrative evidence.

Engineering judgement is based on coverage, reliability, scope compliance and repeatability rather than on the number of tools used. If a result is uncertain, state the uncertainty and request confirmation. If an action would exceed the approved boundary, stop and record the reason instead of continuing in the name of completeness.

Method and evidence checklist

• Confirm written authorization and target scope.

• Record the source, query or test input, timestamp and result.

• Separate prediction, observation and interpretation.

• Mark unverified, stale, duplicate and out-of-scope items.

• Protect evidence and retain only information needed for the assessment.

Evidence / comparison table

Criterion	Measurement	Acceptance rule
Scope	Out-of-scope items enriched	Zero
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Process representation

Select workflow → Diagnose → Define criteria → Redesign → Equivalent test → Compare → Defend

Note: Examples and numerical observations are illustrative academic test data unless identified as command patterns. They are not represented as live collection from a real organization.

Source note

Primary source: ITA14 Ethical Hacking, Assignment 2–CO2 brief supplied for this task, including Q1–Q7, evidence expectations, submission requirements and viva expectations. The brief does not provide live target outputs; this report therefore uses clearly labelled simulated test cases.